Data Mining & Clustering Analysis

Core Foundations: Structured Data Representation, Proximity Metrics & Partitioning Algorithms

Course Synthesis

DM583 Data Mining — Prof. Ricardo J. G. B. Campello

1. Fundamentals of Data Representation

Data mining and exploratory data analysis (EDA) begin with appropriate data structuring. Data can be broadly divided into **structured** (organized in pre-defined formats like tables/relational databases) and **unstructured** (raw text, audio, images). Analysis of unstructured data typically follows a two-stage approach: conversion into a structured matrix representation (e.g., embeddings, bag-of-words, feature vectors), followed by formal structured analysis.

1.1 Taxonomy of Attribute/Variable Types

Tabular data consists of rows representing records/observations/data points (N) and columns representing attributes/variables (n).

Category	Subtype	Description & Examples	Permissible Operations
Categorical *(Qualitative)*	**Nominal**	Values are distinct names/categories with no inherent order (e.g., Car Make, Marital Status, Color). Binary is a special 2-value case.	Equality: $=$, $\neq$
	Ordinal	Qualitative values with a meaningful rank order (e.g.,	Equality & Comparison: $=$

Category	Subtype	Description & Examples	Permissible Operations
		Pain Level $[\text{None} < \text{Mild} < \text{Severe}]$, Education Degree).	$, \neq, <, >$
Numerical *(Quantitative)*	**Discrete**	Finite or countably infinite values, often counts (e.g., Number of children, Word counts). Integer arithmetic applies.	Arithmetic & Rank: $=, \neq, <, >, +, -$
	Continuous	Real-valued measurements on a continuum (e.g., Height, Temperature, Concentration, Elapsed Time).	Full real-number mathematical operations

Symmetric vs. Asymmetric Variables

- **Symmetric Variables:** Presence (1) and absence (0) carry equal importance (e.g., Gender/Sex).
- **Asymmetric Variables:** Only non-zero values (presences) are informative; zeros represent background/absence and dominate sparse datasets (e.g., Word counts in document vectors, Market basket items). Standard distances break down on asymmetric data.

1.2 Specialized Structured Representations

- **Bag-of-Words (BOW) & Text Embeddings:** Text documents are mapped to term-frequency vectors (TF-IDF) or dense machine learning representations (doc2vec, word2vec).
- **Transactions (User-Item Matrix):** Large, sparse matrices recording purchases or ratings.
- **Time-Series Data:** Ordered sequences of real-valued measurements across equal time intervals. Column order is temporally fixed.

- **Proximity Matrices ($D_{N\times N}$):** Square symmetric matrices storing pre-computed pairwise (dis)similarities, enabling privacy-preserving data analysis without revealing original feature values.

2. Proximity & Distance Measures

Proximity reflects either similarity $s(x_i, x_j) \in [0, 1]$ (higher = more alike) or dissimilarity / distance $d(x_i, x_j) \geq 0$ (higher = more different).

Mathematical Metric Requirements for Distance $d(x, y)$

1. **Non-negativity:** $d(x_i, x_j) \geq 0$
2. **Identity of Indiscernibles:** $d(x_i, x_j) = 0 \iff x_i = x_j$
3. **Symmetry:** $d(x_i, x_j) = d(x_j, x_i)$
4. **Triangle Inequality (Metric property):** $d(x_i, x_j) \leq d(x_i, x_k) + d(x_k, x_j)$

2.1 Minkowski Distance Family (Numerical Variables)

For two vectors $x_i, x_j \in \mathbb{R}^n$, the Minkowski distance parameterized by $p \geq 1$ is:

$$d(x_i, x_j) = \|x_i - x_j\|_p = \left(\sum_{k=1}^{n} |x_{ik} - x_{jk}|^p\right)^{1/p}$$

- **Manhattan / City-Block Distance ($p = 1$):**

  $$d(x_i, x_j) = \sum_{k=1}^{n} |x_{ik} - x_{jk}|$$

  Sum of absolute differences along coordinates. Equidistant geometry forms a rhombus/diamond.

- **Euclidean Distance ($p = 2$):**

$$d(x_i, x_j) = \sqrt{\sum_{k=1}^{n} (x_{ik} - x_{jk})^2}$$

Straight-line length (2-norm). Equidistant geometry forms a hyper-sphere.

- **Supremum / Maximum Distance ($p \to \infty$):**

$$d(x_i, x_j) = \max_{1 \le k \le n} |x_{ik} - x_{jk}|$$

Maximum single coordinate difference. Equidistant geometry forms a hyper-cube.

2.2 Mahalanobis Distance (Covariance-Aware Metric)

Measures distance from a point x to a cluster/distribution center v while accounting for variable variance and linear inter-correlations:

$$d_{\text{Mahalanobis}}(x, v) = \sqrt{(x - v)^T \text{Cov}(X)^{-1} (x - v)}$$

Where $\text{Cov}(X)$ is the $n \times n$ sample covariance matrix. Unlike Euclidean distance, Mahalanobis induces ellipsoidal equidistant contours, making it scale-invariant and resilient to correlated attributes.

2.3 Variable Standardization & Rescaling

Distance measures are highly sensitive to variable scales. Attributes with large ranges artificially dominate distance calculations.

- **Linear Min-Max Rescaling $[0, 1]$:**

$$y_{ij} = \frac{x_{ij} - \min(a_j)}{\max(a_j) - \min(a_j)}$$

- **Z-Score Standardization:**

$$z_{ij} = \frac{x_{ij} - \mu_{a_j}}{\sigma_{a_j}}$$

2.4 Trend-Based & Sparse Similarity Measures

- **Pearson Correlation Coefficient (r):** Measures linear trend similarity across attributes ($r \in [-1, +1]$). Highly sensitive to extreme values/outliers.

$$r(x_i, x_j) = \frac{\mathrm{cov}(x_i, x_j)}{\sigma_{x_i} \cdot \sigma_{x_j}}$$

- **Spearman Rank Correlation (r_s):** Replaces feature values with their relative ranks before computing Pearson. Robust to outliers and non-linear monotonic relationships.

- **Cosine Similarity:** Ideal for sparse/asymmetric numerical data (e.g., BOW text mining). Computes the cosine of the angle between vectors, ignoring shared zero attributes:

$$\cos(x_i, x_j) = \frac{\langle x_i, x_j \rangle}{\|x_i\|_2 \cdot \|x_j\|_2} = \frac{\sum_{k=1}^{n} x_{ik} x_{jk}}{\sqrt{\sum_{k=1}^{n} x_{ik}^2} \cdot \sqrt{\sum_{k=1}^{n} x_{jk}^2}}$$

2.5 Proximity for Binary Categorical Attributes

Given two binary vectors, compute the contingency table counts: n_{11} (both 1), n_{00} (both 0), n_{10} ($x_i = 1, x_j = 0$), n_{01} ($x_i = 0, x_j = 1$).

- **Simple Matching Coefficient (SMC):** Used for symmetric binary variables (0-0 matches matter).

$$\mathrm{SMC} = \frac{n_{11} + n_{00}}{n_{11} + n_{00} + n_{10} + n_{01}} = \frac{n_{11} + n_{00}}{n}$$

- **Jaccard Coefficient (J_c):** Used for asymmetric binary variables (ignores 0-0 matches).

$$J_c = \frac{n_{11}}{n_{11} + n_{10} + n_{01}}$$

3. Clustering Analysis & Partitioning Methods

Clustering is an exploratory data analysis technique aimed at grouping unlabeled objects such that internal cohesion (homogeneity) is maximized and external separation (heterogeneity) is maximized.

3.1 Mathematical Definition of Hard Data Partitioning

Given dataset $X = \{x_1, x_2, \ldots, x_N\}$, a hard (non-overlapping) partition $\mathcal{C} = \{C_1, C_2, \ldots, C_k\}$ into k clusters satisfies:

1. $\bigcup_{c=1}^{k} C_c = X$ (Exhaustive covering)
2. $C_c \neq \emptyset \quad \forall c \in \{1, \ldots, k\}$ (No empty clusters)
3. $C_i \cap C_j = \emptyset \quad \forall i \neq j$ (Mutually disjoint)

A partition can be represented by a Partition Matrix $U_{k \times N}$ with binary membership entries $\mu_{cj} \in \{0, 1\}$ where $\sum_{c=1}^{k} \mu_{cj} = 1, \quad \forall j$.

3.2 K-Means Clustering Algorithm

K-Means is a prototype-based partitioning algorithm for real-valued quantitative variables using squared Euclidean distance. It aims to minimize the Sum of Squared Errors (SSE) / Within-Cluster Variance:

$$J = \mathrm{SSE} = \sum_{c=1}^{k} \sum_{x_j \in C_c} \|x_j - \bar{x}_c\|_2^2 = \sum_{j=1}^{N} \sum_{c=1}^{k} \mu_{cj} \|x_j - \bar{x}_c\|_2^2$$

where $\bar{x}_c = \frac{1}{|C_c|} \sum_{x_j \in C_c} x_j$ is the centroid (mean vector) of cluster C_c.

Iterative Alternating Optimization (Expectation-Maximization)

1. **Initialization (Step 1):** Choose k initial cluster prototypes.
2. **Assignment / E-step (Step 2):** Fix centroids $\{\bar{x}_c\}$, minimize J w.r.t. memberships μ_{cj} by assigning each observation to its closest centroid:

$$\mu_{cj} = 1 \quad \text{for } c = \arg\min_{l} \|x_j - \bar{x}_l\|^2$$

3. **Update / M-step (Step 3):** Fix memberships μ_{cj}, set gradient $\nabla_{\bar{x}_c} J = 0$ to recalculate centroids as cluster arithmetic means:

$$\bar{x}_c = \frac{\sum_{j=1}^{N} \mu_{cj} x_j}{\sum_{j=1}^{N} \mu_{cj}}$$

4. **Convergence (Step 4):** Repeat Steps 2 and 3 until cluster assignments stabilize or max iterations are reached.

Critical Guarantee & Limitations: K-Means is guaranteed to converge to a local optimum, but NOT necessarily the global minimum. Results are sensitive to initial prototype selection.

3.3 Algorithmic Optimizations & Scalability

- **K-Means++ Initialization:** Chooses the 1st center uniformly at random, and subsequent centers from remaining data points with probability proportional to their squared distance to the nearest existing center ($D(x)^2$). Significantly improves global convergence quality.

- **Recursive Centroid Updates:** When an object x moves from cluster C_A to C_B, updated centroids can be computed directly without full recomputation:

$$\bar{x}_A^{\text{new}} = \bar{x}_A^{\text{old}} - \frac{x - \bar{x}_A^{\text{old}}}{|C_A| - 1}, \qquad \bar{x}_B^{\text{new}} = \bar{x}_B^{\text{old}} + \frac{x - \bar{x}_B^{\text{old}}}{|C_B| + 1}$$

- **Parallel / Distributed K-Means:** Data is partitioned across P processors. Each node computes local cluster counts and sum vectors in parallel. A master node aggregates local counts to compute exact global centroids and broadcasts them back.

- **Acceleration Structures:** KD-trees and exploitation of the Triangle Inequality skip redundant distance computations during assignment steps.

3.4 Determining Hyperparameter k & Cluster Validation

Elbow Method (SSE Curve)	Silhouette Width Criterion (SWC)

Elbow Method (SSE Curve)	Silhouette Width Criterion (SWC)
Plots SSE vs. k across multiple runs. SSE decreases monotonically as k increases (reaching $\mathrm{SSE} = 0$ at $k = N$). An "elbow" point indicates diminishing returns in variance reduction.	Measures how well-clustered each observation is ($s(i) \in [-1, +1]$): $$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$ where $a(i)$ is mean intra-cluster dist, and $b(i)$ is mean dist to nearest neighboring cluster. Simplified version uses centroid distances ($O(Nk)$ runtime vs. original $O(N^2)$).

3.5 Summary Assessment of K-Means

Strengths / Pros	Weaknesses / Cons
• Simple, intuitive, and highly interpretable.	• Requires prior specification of hyperparameter k.
• Computationally fast with linear complexity $O(N \cdot n \cdot k \cdot I)$.	• Sensitive to initialization (prone to sub-optimal local minima).
• Scales well to large datasets via parallel execution.	• Assumes convex, spherical/globular clusters of equal variance.
	• Sensitive to outliers (mean statistic is non-robust).

Strengths / Pros	Weaknesses / Cons
	• Restricted to real-valued quantitative variables.